

Beyond the Square-Root Cusp

Logarithmic critical corrections and equilibrium
selection for digital-product pressure

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Abstract

The ProveIt critical-pressure article determines a square-root singularity and explicitly asks for its next term and for the limiting equilibrium measures. We address both questions. For the consecutive digital mask in any integer base $b \geq 2$, put $\kappa_b = \sqrt{2(b-1)\log b}$. At the critical absolute-moment exponent, the unnormalized cylinder pressure is

$$P_{b,1}(c) = \kappa_b \sqrt{|c|} - (b-1)|c| \log(1/|c|) \\ + [b \log b + (b-1)(\gamma-1)]|c| + O_{b,\eta}(|c|^{3/2-\eta})$$

for every $0 < \eta < 1/2$. Thus both a logarithmic term and an ordinary linear term occur. The logarithm comes from an explicitly evaluated finite-part response functional, rather than from a formal continuation of a supercritical zeta coefficient.

At nonzero phases near zero there is a unique equilibrium probability. At the critical exponent it converges weakly to $\frac{1}{2}\delta_0 + \frac{1}{2}\mathfrak{m}$, where $\mathfrak{m}$ is normalized Lebesgue measure. More generally, along $s = 1 + u\sqrt{|c|}$, bounded detuning selects an explicit mixture whose atomic mass is

$$\frac{1}{2} \left(1 + \frac{u \log b}{\sqrt{u^2 (\log b)^2 + 8(b-1) \log b}} \right).$$

We obtain the individual left and right eigenvector asymptotics, a $\sqrt{|c|}$ -scale boundary layer, and the exact leading divergence of a finite-product amplitude. The spectral foundation and the leading crossover are credited to the repository predecessor and reconstructed as needed. Detailed proofs, reproducible diagnostics, a claims audit, and further research questions accompany the article.

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1 The precise research target

1.1 What was already known in the inspected repository

The immediate predecessor is *Critical Cusps in Digital-Product Pressure* [4]. It proves the leading law $P_{b,1}(c) \sim \kappa_b \sqrt{|c|}$, constructs a size-two Jordan block at the atomic phase, and obtains the leading bounded-detuning and finite-size crossover. Those results are *not* claimed anew here. Its further research agenda contains two particularly concrete questions:

1. After the square root, does the pressure have a $|c| \log |c|$ term, a $|c|$ term, or both, and what are their coefficients?
2. Which normalized eigenvectors and equilibrium measures emerge as the phase tends to the atomic phase?

These are the predecessor’s questions entitled “The next critical term” and “Eigenfunctions, eigenmeasures, and equilibrium measures.” The present article answers the first with an explicit remainder and answers the weak-selection and first boundary-layer parts of the second.

The earlier *Fractional Cusps at the Atomic Phase* treats exponents $s > 1$ [5]. A later subcritical article treats all $0 < s < 1$ [6]. Neither the leading square root nor the subcritical linear response is a new contribution of this package. This distinction matters in a rapidly expanding research corpus.

The primary literature studies generalized Thue–Morse measures and phase-dependent singular potentials [1, 2]. In particular, [2, Theorems 2.2–2.3 and the following discussion] proves phase continuity at nonnegative pressure orders and asks about stronger regularity at general positive orders. Our result is a refined local answer at the exceptional atomic phase; it is not a global regularity classification. The published sine convention is obtained from our binary cosine convention by shifting the phase by $1/2$.

1.2 Claims and boundaries

Status	Content
Prior repository results	Leading square-root pressure, the atomic Jordan block, and the leading bounded-detuning eigenvalue crossover.
Proved here	Both next critical coefficients with an explicit little-power-loss remainder; first normalized eigenvector corrections; critical and detuned equilibrium selection; the right boundary layer.
Derived consequences	An exact amplitude constant and the obstruction to a pure half-integer-power expansion through order $ c $.
Not established	A complete transseries, a sharp next coefficient, unbounded-detuning uniformity, a global phase theory, or formal verification.

All model-specific analytic ingredients used below are proved or reconstructed. We use the standard elementary framework of bounded operators, Riesz projections, compact approximation, and measure entropy; the particular finite-dimensional reduction and entropy argument are spelled out. Numerical calculations are a separate diagnostic layer.

The pinned repository snapshot is

1085b506d65e207a05b7e9c861bb1fe88432fe38.

The inspected critical article has blob identifier 5539ac92693fc99fa35ded81bcf9a4486d37e501. The cited primary papers and targeted repository searches establish a specific local research gap, not worldwide priority. Independent review is still needed. No repository files were changed in preparing this work.

2 Definitions and main results

2.1 Normalization

Fix $b \geq 2$ and let $T_b x = bx \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Write

$$m_b(x) = \left| \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r x} \right| = \left| \frac{\sin \pi b x}{b \sin \pi x} \right|, \quad (2.1)$$

with removable values at integers. For $s > 0$ and phase c , define

$$(\mathcal{L}_{s,c} f)(x) = \sum_{k=0}^{b-1} m_b \left(\frac{x+k}{b} - c \right)^s f \left(\frac{x+k}{b} \right). \quad (2.2)$$

There is no factor $1/b$ in this operator. Let $\mathcal{D}_{b,n}$ be the b^n equal closed cells of $[0, 1]$. Put

$$F_{s,n,c}(x) = \prod_{j=0}^{n-1} m_b(b^j x - c)^s, \quad (2.3)$$

$$Z_{s,n}(c) = \sum_{I \in \mathcal{D}_{b,n}} \sup_{x \in I} F_{s,n,c}(x), \quad (2.4)$$

$$P_{b,s}(c) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{s,n}(c). \quad (2.5)$$

The usual subdivision inequality makes $Z_{s,n}$ submultiplicative; in the neighborhoods used here we will also identify the limit directly with a simple eigenvalue. All logarithms are natural. Throughout,

$$\delta = |c|, \quad t = \sqrt{\delta}, \quad d = b - 1, \quad L = \log b, \quad a = 2L, \quad \kappa_b = \sqrt{ad}. \quad (2.6)$$

Euler's constant is γ , and $\mathfrak{m}$ denotes normalized Lebesgue measure on $\mathbb{T}$. The binary pressure order is

$$P(q \log \cos^2 \pi(x - c)) = P_{2,2q}(c). \quad (2.7)$$

Thus our critical exponent $s = 1$ corresponds to $q = 1/2$.

2.2 The complete expansion through the linear scale

Theorem 2.1 (Logarithmic critical correction). *For every integer $b \geq 2$ and every $0 < \eta < 1/2$, as $c \rightarrow 0$,*

$$\boxed{P_{b,1}(c) = \kappa_b \sqrt{\delta} - d\delta \log(1/\delta) + B_b \delta + O_{b,\eta}(\delta^{3/2-\eta}),} \quad (2.8)$$

$$B_b = b \log b + (b-1)(\gamma-1).$$

For small nonzero c , the two eigenvalues near one satisfy

$$\lambda_{\pm}(c) = 1 \pm \kappa_b \sqrt{\delta} - d\delta \log(1/\delta) + C_b \delta + O_{b,\eta}(\delta^{3/2-\eta}), \quad (2.9)$$

where $C_b = (2b-1) \log b + (b-1)(\gamma-1)$. The remaining spectrum on a suitable Hölder space stays in a fixed disk of radius less than one, and $P_{b,1}(c) = \log \lambda_+(c)$.

For example,

$$P_{2,1}(c) = \sqrt{2 \log 2} \sqrt{\delta} - \delta \log(1/\delta) + (2 \log 2 + \gamma - 1) \delta + O_{\eta}(\delta^{3/2-\eta}). \quad (2.10)$$

Here $B_2 \approx 0.9635100260$. A polynomial in $\sqrt{\delta}$ cannot represent the expansion through order δ : the nonzero $\delta \log \delta$ term must be retained. This is an obstruction to a pure Puiseux expansion, not a proof of a complete logarithmic transseries.

2.3 Eigenvectors, boundary layers, and selected measures

Define, for $0 < x < 1$,

$$H(x) = \frac{\sin \pi x}{\pi}, \quad G(x) = H(x) \left[-\psi(x) - \psi(1-x) + 2 \log \frac{2}{\pi} \right], \quad (2.11)$$

where $\psi = \Gamma'/\Gamma$. Extend periodically with $H(0) = 0$ and $G(0) = 1$. For $f \in C^\alpha$, $0 < \alpha < 1$, set

$$\ell(f) = f(0), \quad \Xi(f) = \int_0^1 \frac{f(x) - f(0)}{H(x)} dx. \quad (2.12)$$

The integral is absolutely convergent. The functional Ξ is bounded on C^α but is not asserted to be a finite signed measure.

Theorem 2.2 (Critical eigenvectors and half-and-half selection). *Fix $0 < \alpha < 1/2$. For sufficiently small $c \neq 0$ there are a strictly positive right eigenfunction h_c and a positive left eigenprobability ν_c for $\lambda_+(c)$, uniquely normalized by $\Xi(h_c) = 1$ and $\nu_c(1) = 1$. As $c \rightarrow 0$,*

$$h_c = H + \frac{\kappa_b}{a} \sqrt{\delta} G + o(\sqrt{\delta}) \quad \text{in } C^\alpha, \quad (2.13)$$

$$\nu_c = \ell + \frac{\kappa_b}{a} \sqrt{\delta} \Xi + o(\sqrt{\delta}) \quad \text{in } (C^\alpha)^*. \quad (2.14)$$

The right boundary layer is

$$\frac{h_c(\sqrt{\delta} z)}{\sqrt{\delta}} \longrightarrow |z| + \frac{\kappa_b}{a}, \quad (2.15)$$

uniformly for z in each compact subset of $\mathbb{R}$, with the argument interpreted on the circle. The probability

$$\mu_c(f) = \frac{\nu_c(h_c f)}{\nu_c(h_c)} \quad (2.16)$$

is the unique equilibrium measure for $\log m_b(x - c)$, and

$$\boxed{\mu_c \implies \frac{1}{2} \delta_0 + \frac{1}{2} \mathbf{m}.} \quad (2.17)$$

The arrow means weak convergence of probabilities, not convergence in total variation.

Theorem 2.3 (Bounded-detuning selection). *Let $c = \sigma t^2$, $s = 1 + ut$, where $\sigma \in \{-1, 1\}$ and $t \downarrow 0$. For u in any fixed compact subset of $\mathbb{R}$, the unique equilibrium probabilities of $s \log m_b(x - c)$ satisfy, uniformly against each continuous test function,*

$$\boxed{\mu_{1+ut, \sigma t^2} \implies w_b(u) \delta_0 + (1 - w_b(u)) \mathbf{m}, \quad w_b(u) = \frac{1}{2} \left(1 + \frac{uL}{\sqrt{u^2 L^2 + 4\kappa_b^2}} \right).} \quad (2.18)$$

At $(s, c) = (1, 0)$ the entire set of equilibrium probabilities is exactly $\{v \delta_0 + (1 - v) \mathbf{m} : 0 \leq v \leq 1\}$. Thus the phase perturbation and its detuning select particular elements of a pre-existing equilibrium segment.

The limiting weights tend algebraically to one or zero as $u \rightarrow +\infty$ or $u \rightarrow -\infty$. The theorem does not make its error uniform along an unbounded function $u = u(t)$.

3 Reconstructing the critical spectral foundation

The Jordan mechanism in this section is a predecessor result [4]. We reconstruct it to fix the normalization of G and the dual coordinates, which is essential for the next coefficients. Our G differs from the predecessor's unnormalized digamma companion by a constant multiple of H .

3.1 An exact Jordan chain and its dual

Write $\mathcal{L} = \mathcal{L}_{1,0}$. The mask identity gives

$$(\mathcal{L}f)(x) = \frac{H(x)}{b} \sum_{k=0}^{b-1} \frac{f((x+k)/b)}{H((x+k)/b)}, \quad 0 < x < 1. \quad (3.1)$$

The removable endpoint values are understood by continuity.

Lemma 3.1 (Normalized Jordan coordinates). *The functions and functionals in equations (2.11) and (2.12) satisfy*

$$\mathcal{L}H = H, \quad \mathcal{L}G = G + aH, \quad \ell\mathcal{L} = \ell, \quad \Xi\mathcal{L} = \Xi + a\ell, \quad (3.2)$$

$$\ell(G) = 1, \quad \ell(H) = 0, \quad \Xi(H) = 1, \quad \Xi(G) = 0. \quad (3.3)$$

Both G and H are periodic Lipschitz functions. Consequently

$$\Pi f = G\ell(f) + H\Xi(f), \quad \mathcal{N}f = aH\ell(f) \quad (3.4)$$

define a bounded rank-two projection and a nonzero nilpotent operator with $\mathcal{N}^2 = 0$ and $\mathcal{L}\Pi = \Pi\mathcal{L} = \Pi + \mathcal{N}$.

Proof. The expansions $\psi(x) = -x^{-1} - \gamma + O(x)$ and $\psi(1-x) = -\gamma + O(x)$ give $G(0) = 1$ with a finite one-sided derivative; reflection handles the other endpoint. Formula equation (3.1) immediately fixes H . The digamma multiplication identity

$$\sum_{k=0}^{b-1} \psi\left(\frac{x+k}{b}\right) = b\psi(x) - b\log b$$

and its reflected version show that $\mathcal{L}G = G + 2\log b H$. The multiplication identity follows by logarithmic differentiation of the Gamma multiplication formula, or directly by grouping the convergent series for differences of digamma functions into residue classes.

For the only nontrivial normalization, integrate on $[\varepsilon, 1-\varepsilon]$. Since $\psi = (\log \Gamma)'$,

$$\int_{\varepsilon}^{1-\varepsilon} [-\psi(x) - \psi(1-x)] dx = 2\log \Gamma(\varepsilon) - 2\log \Gamma(1-\varepsilon),$$

$$\int_{\varepsilon}^{1-\varepsilon} \frac{dx}{H(x)} = 2\log \cot(\pi\varepsilon/2).$$

Using $\log \Gamma(\varepsilon) = -\log \varepsilon + O(\varepsilon)$, the constant $2\log(2/\pi)$ in G/H cancels the remaining finite part. This proves $\Xi(G) = 0$.

If $f(0) = 0$, equation (3.1) and a change of variables give $\Xi(\mathcal{L}f) = \Xi(f)$; absolute integrability follows from $|f(x)| \leq C \operatorname{dist}(x, \mathbb{Z})^\alpha$. For arbitrary f , subtract $f(0)G$ and use the already proved formula for $\mathcal{L}G$. This proves the dual identity. The projection and commutation assertions are now direct consequences of the four pairings in equation (3.3). $\square$

3.2 Weighted quadrature and the complementary spectrum

Lemma 3.2 (Uniform iterate estimate). *For $0 < \alpha < 1$, $N = b^n$, and $f \in C^\alpha$, one has*

$$\|\mathcal{L}^n f - anHf(0) - Gf(0) - H\Xi(f)\|_\infty \leq C_{b,\alpha}(n+1)b^{-\alpha n}\|f\|_\alpha. \quad (3.5)$$

Proof. Iterating the telescoping quotient in equation (3.1) yields

$$\mathcal{L}^n f(x) = \frac{H(x)}{N} \sum_{k=0}^{N-1} \frac{f((x+k)/N)}{H((x+k)/N)}. \quad (3.6)$$

First let $f = 1$. Subtract the two endpoint poles from the cosecant:

$$v(y) = \csc \pi y - \frac{1}{\pi y} - \frac{1}{\pi(1-y)}.$$

This extends smoothly to $[0, 1]$, and direct integration gives $\int_0^1 v = (2/\pi) \log(2/\pi)$. The two pole sums are evaluated using

$$\sum_{k=0}^{N-1} \frac{1}{x+k} = \psi(N+x) - \psi(x).$$

Since $\psi(N+x) = \log N + O(N^{-1})$ uniformly for $0 \leq x \leq 1$, and $H(x)$ removes the apparent endpoint divergences, the resulting expression is $2 \log N H(x) + G(x) + O(N^{-1})$, uniformly in x .

Now put $g = f - f(0)$. The function g/H is integrable, although generally unbounded. Compare the shifted sample in each cell of length $1/N$ with its integral. For the first cell, the weighted sample is bounded by

$$\frac{H(x)}{N} \frac{|g(x/N)|}{H(x/N)} \leq C \|f\|_\alpha N^{-\alpha} H(x) x^{\alpha-1} \leq C' \|f\|_\alpha N^{-\alpha};$$

the last cell is analogous. The endpoint integrals have the same bound. For an interior cell at distance comparable to k/N from an endpoint, the variation of the numerator contributes at most $C \|f\|_\alpha N^{-\alpha} / k$ after integration; the variation of $1/H$ contributes at most $C \|f\|_\alpha N^{-\alpha} k^{\alpha-2}$. The same estimates hold with $N - k$ near the other endpoint. Their sums are $O(N^{-\alpha} \log N)$ and $O(N^{-\alpha})$, respectively. Thus the weighted sum converges uniformly to $H(x)\Xi(f)$ at the stated rate. The case $n = 0$ is absorbed by increasing the constant. $\square$

Proposition 3.3 (Spectral separation). *On C^α , the spectral subspace of $\mathcal{L}$ at eigenvalue one is exactly $\text{span}\{G, H\}$, with its size-two Jordan block. All remaining spectrum has modulus at most $b^{-\alpha}$. In particular, for any $b^{-\alpha} < \theta < 1$ there is $K_\theta < \infty$ such that*

$$\|\mathcal{L}^n - \Pi - n\mathcal{N}\|_{\alpha \rightarrow \alpha} \leq K_\theta \theta^n \quad (n \geq 1). \quad (3.7)$$

Proof. For each fixed n , the inverse-branch formula and the product inequality for Hölder seminorms give a Lasota–Yorke estimate

$$[\mathcal{L}^n f]_\alpha \leq C b^{-\alpha n} \|\mathcal{L}^n 1\|_\infty [f]_\alpha + C_n \|f\|_\infty. \quad (3.8)$$

Here C is independent of n . On arcs crossing the coordinate cut, reindex the inverse branches and split at the cut; this changes only the fixed Hölder constant. The coefficients are Lipschitz for each n .

To see the compactness consequence explicitly, approximate the identity by periodic piecewise-linear interpolation Q_M on a uniform mesh. These finite-rank maps have uniformly bounded C^α norms and $\|I - Q_M\|_{\alpha \rightarrow 0} = O(M^{-\alpha})$. Apply equation (3.8) to $(I - Q_M)f$ and let $M \rightarrow \infty$. The essential norm of $\mathcal{L}^n$ is at most $C' b^{-\alpha n} \|\mathcal{L}^n 1\|_\infty$. By lemma 3.2, the last factor is $O(n+1)$, so $r_{\text{ess}}(\mathcal{L}) \leq b^{-\alpha}$.

Outside that disk, spectrum consists of isolated eigenvalues of finite algebraic multiplicity. If $\mathcal{L}f = \lambda f$ and $\lambda \neq 1$, the dual relations imply $\ell(f) = \Xi(f) = 0$. Formula equation (3.5) then excludes $|\lambda| > b^{-\alpha}$. A generalized eigenvector at one, after subtracting its Π component, also lies in the common kernel of the two functionals. Its iterates are a polynomial in n , whereas equation (3.5) makes them tend to zero. That polynomial must vanish identically. The entire generalized eigenspace is therefore the range of Π . On its complement the spectral-radius formula gives equation (3.7), since the central block has exact powers $\Pi + n\mathcal{N}$. $\square$

4 A perturbation reduction with a controlled chart error

4.1 The strength of the phase perturbation

For the remainder of the critical calculation, let $E_c = \mathcal{L}_{1,c} - \mathcal{L}_{1,0}$. The absolute trigonometric mask is Lipschitz, so its translated difference has uniform norm $O(\delta)$ and bounded Lipschitz seminorm. Splitting a Hölder quotient into distances below and above δ proves

$$\|E_c\|_{\alpha \rightarrow \alpha} = O(\delta^{1-\alpha}) = O(\varepsilon_c), \quad \varepsilon_c = \delta^{1-\alpha}. \quad (4.1)$$

The finite inverse-branch sum is bounded on C^α , so the weight estimate indeed gives the operator estimate. It does not assert norm differentiability of translation at a cusp.

4.2 Graph coordinates around an isolated spectral cluster

Lemma 4.1 (Second-order compression error). *Let A_0 be a bounded operator with an isolated finite-dimensional spectral subspace V and commuting projection P . Put $Q = I - P$. For sufficiently small bounded E , the corresponding spectral subspace of $A_0 + E$ is the graph of a map $K : V \rightarrow QX$ with $\|K\| = O(\|E\|)$. In the graph coordinate $v \mapsto v + Kv$, the compressed operator is*

$$A_0|_V + PE|_V + O(\|E\|^2). \quad (4.2)$$

The constants are uniform in a continuous family with a common separating contour and uniformly bounded resolvents.

Proof. A resolvent Neumann series on the separating contour shows that the Riesz projection P_E satisfies $\|P_E - P\| = O(\|E\|)$. The restriction of P to $\text{ran } P_E$ is invertible onto V , so that range is a graph, with $K = O(\|E\|)$. Projection back to V gives the exact compressed expression

$$P(A_0 + E)(I + K)|_V = A_0|_V + PE|_V + PEK.$$

The term PA_0K vanishes because QX is A_0 -invariant. The final term has norm $O(\|E\|^2)$. This proves the claim without selecting individual eigenvectors inside a possibly defective cluster. $\square$

In the ordered basis (G, H) , lemma 4.1 and proposition 3.3 give the two-dimensional matrix

$$A_c = \begin{pmatrix} 1 + \ell(E_c G) & \ell(E_c H) \\ a + \Xi(E_c G) & 1 + \Xi(E_c H) \end{pmatrix} + O(\delta^{2-2\alpha}). \quad (4.3)$$

The error is an entrywise estimate in a fixed basis. It is not discarded without examining its effect on the square-root splitting: an $O(\delta^{2-2\alpha})$ error in the small upper-right entry can shift an eigenvalue by $O(\delta^{3/2-2\alpha})$. Choosing $\alpha < 1/4$ is what makes this $o(\delta)$ and preserves the linear coefficient.

5 Exact branch identities and the finite-part logarithm

Reflection conjugates $\mathcal{L}_{1,c}$ to $\mathcal{L}_{1,-c}$, so eigenvalues and pressure are even. We therefore take $0 < c < 1/(4b)$ in this section and restore $\delta = |c|$ at the end.

5.1 A signed cotangent calculation

Lemma 5.1 (Exact response of the vanishing eigenfunction). *For $d = b - 1$,*

$$(E_c H)(0) = \frac{\sin \pi d c}{\pi}. \quad (5.1)$$

On $bc \leq x \leq 1$,

$$(\mathcal{L}_{1,c}H)(x) = \frac{\sin \pi(x - dc)}{\pi}. \quad (5.2)$$

On $0 \leq x \leq bc$,

$$(\mathcal{L}_{1,c}H)(x) = 2m_b(x/b - c)H(x/b) - \frac{\sin \pi(x - dc)}{\pi}. \quad (5.3)$$

Proof. Set $y_k = (x+k)/b$. In the numerator absolute value, absorb the factor $(-1)^k$ from $\sin \pi b(y_k - c)$, leaving $|\sin \pi(x - bc)|$. Introduce the auxiliary signed sum obtained from this expression by omitting its two absolute values. After multiplication by $H(y_k)$ it is

$$\frac{\sin \pi(x - bc)}{b\pi} \sum_{k=0}^{b-1} \frac{\sin \pi y_k}{\sin \pi(y_k - c)}.$$

Use $\sin \pi y_k / \sin \pi(y_k - c) = \cos \pi c + \sin \pi c \cot \pi(y_k - c)$ and

$$\sum_{k=0}^{b-1} \cot \pi \left(\frac{x+k}{b} - c \right) = b \cot \pi(x - bc).$$

The sum becomes $\sin \pi(x - dc)/\pi$. When $x \geq bc$, both the common numerator and all denominators have the sign yielding the absolute-value sum, so this proves equation (5.2). When $0 < x < bc$, the branch $k = 0$ has negative numerator and denominator, whereas the other denominators remain positive. Restoring absolute values reverses the other branch signs. The total is twice the zeroth-branch contribution minus the signed sum, giving equation (5.3). Continuity at $x = 0$, where $H(0) = 0$, gives equation (5.1). Apparent cotangent singularities are removed by their sine prefactor. $\square$

5.2 The three matrix entries that determine the answer

Proposition 5.2 (Explicit response coefficients). *As $c \downarrow 0$,*

$$\ell(E_c H) = dc + O(c^3), \quad (5.4)$$

$$\ell(E_c G) = r_b c + O(c^2), \quad (5.5)$$

$$\Xi(E_c H) = -2dc \log(1/c) + k_b c + O(c^2), \quad (5.6)$$

where

$$r_b = 2d \left(\gamma + \log \frac{2}{\pi} \right) + 2b \log b, \quad k_b = 2d \left(\log \frac{\pi b}{2} - 1 \right). \quad (5.7)$$

For the remaining entry, the bound $\Xi(E_c G) = O(c^{1-\alpha})$ suffices.

Proof. The first line is equation (5.1). Evaluating $\mathcal{L}_{1,c}G$ at zero, the branch at zero changes by $O(c^2)$ since $m_b(-c) = 1 + O(c^2)$. For $1 \leq j < b$,

$$m_b(j/b - c) = \frac{\pi c}{\sin(\pi j/b)} + O(c^2) = \frac{c}{H(j/b)} + O(c^2).$$

It follows that the coefficient in equation (5.5) is $\sum_{j=1}^{b-1} G(j/b)/H(j/b)$. The digamma multiplication identity gives

$$\sum_{j=1}^{b-1} \psi(j/b) = -d\gamma - b \log b,$$

which evaluates the coefficient as r_b .

For the logarithmic entry, write $v = \pi dc$. By definition,

$$\Xi(E_c H) = \int_0^1 \frac{\mathcal{L}_{1,c} H(x) - H(x) - \sin(v)/\pi}{H(x)} dx. \quad (5.8)$$

This is an ordinary convergent integral: the constant subtracted in the numerator is the value of $E_c H$ at zero. Split it at $x = bc$. On the outer interval, equation (5.2) gives

$$\frac{\mathcal{L}_{1,c} H - H - \sin(v)/\pi}{H} = \cos v - 1 - \sin v [\cot \pi x + \csc \pi x].$$

Since $\cot \pi x + \csc \pi x = \cot(\pi x/2)$, its integral is exactly

$$I_{\text{out}} = (1 - bc)(\cos v - 1) + \frac{2 \sin v}{\pi} \log \sin \frac{\pi bc}{2}. \quad (5.9)$$

Thus

$$I_{\text{out}} = -2dc \log(1/c) + 2dc \log(\pi b/2) + O(c^2). \quad (5.10)$$

On the inner interval, equation (5.3) and Taylor expansion give, uniformly for $0 \leq x \leq bc$ after removing the value at $x = 0$,

$$\frac{\mathcal{L}_{1,c} H - H - \sin(v)/\pi}{H(x)} = -\frac{2d}{b} + O(c^2). \quad (5.11)$$

For a precise uniform justification, $m_b(x/b - c) = 1 + O(c^2)$ there, $H(x/b) = x/b + O(x^3)$, and

$$\sin \pi(x - dc) + \sin \pi dc = \pi x + O(c^2 x).$$

The numerator in equation (5.11) is consequently $-2dx/b + O(c^2 x)$, and division by $H(x) = x(1 + O(c^2))$ is uniform down to the removable endpoint. Hence $I_{\text{in}} = -2dc + O(c^3)$. Combining this with equation (5.10) proves equation (5.6). The last bound follows from boundedness of Ξ and equation (4.1). $\square$

Corollary 5.3 (An exact binary check). *For $b = 2$ and $0 < c < 1/8$,*

$$\Xi(E_c H) = -2c + (1 - 2c)(\cos \pi c - 1) + \frac{2 \sin \pi c}{\pi} \log \sin \pi c. \quad (5.12)$$

Proof. For the binary mask, the inner formula simplifies by elementary sine addition to $\mathcal{L}_{1,c} H(x) = \sin \pi c/\pi$ when $0 \leq x \leq 2c$. The integrand in equation (5.8) is therefore exactly -1 on that interval. Add its integral $-2c$ to equation (5.9). $\square$

The appearance of $\log c$ is now transparent. The shifted zeros feed a nonzero value into a function which previously vanished at zero. The dual coordinate removes that value but integrates the residual against $1/H(x) \sim 1/\text{dist}(x, \mathbb{Z})$. The outer region beginning at bc therefore leaves a logarithmic finite part. No divergent expression has been treated as a convergent integral.

6 From the response matrix to the pressure expansion

6.1 The refined eigenvalue calculation

Proof of the eigenvalue assertion in theorem 2.1. Fix $0 < \eta < 1/2$ and choose $\alpha = \eta/2 < 1/4$. Write the matrix in equation (4.3) as

$$A_c = \begin{pmatrix} 1 + e & u \\ a + g & 1 + f \end{pmatrix}.$$

Its entries, including the graph error, satisfy

$$e = r_b c + O(c^{2-2\alpha}), \quad f = -2dc \log(1/c) + k_b c + O(c^{2-2\alpha}), \quad (6.1)$$

$$u = dc + O(c^{2-2\alpha}), \quad g = O(c^{1-\alpha}). \quad (6.2)$$

The two eigenvalues are exactly

$$1 + \frac{e+f}{2} \pm \sqrt{u(a+g) + \frac{(e-f)^2}{4}}. \quad (6.3)$$

The expression under the square root is

$$adc + O(c^{2-2\alpha}) + O(c^{2-\alpha}) + O(c^2 \log^2 c). \quad (6.4)$$

It is positive for sufficiently small $c > 0$. Dividing the error by $\sqrt{c}$ in the difference-of-square-roots identity gives

$$\sqrt{u(a+g) + (e-f)^2/4} = \kappa_b \sqrt{c} + O(c^{3/2-\eta}). \quad (6.5)$$

Here $c^{3/2} \log^2 c = O_\eta(c^{3/2-\eta})$.

The half-trace is

$$\frac{e+f}{2} = -dc \log(1/c) + \frac{r_b + k_b}{2} c + O(c^{2-2\alpha}). \quad (6.6)$$

The constants simplify to

$$\frac{r_b + k_b}{2} = d(\gamma - 1) + (2b - 1) \log b = C_b. \quad (6.7)$$

Combining these equations proves [equation \(2.9\)](#) for positive phases. Reflection supplies the negative phases. The rank-two cluster is the only spectrum near one; the complementary spectrum stays in a fixed smaller disk by resolvent continuity. The eigenvalues are real, simple, and separated by $2\kappa_b \sqrt{\delta} + o(\sqrt{\delta})$. $\square$

6.2 Positivity and identification with cylinder pressure

Lemma 6.1 (Positive spectral data and pressure). *For sufficiently small $c \neq 0$, $\lambda_+(c)$ is the simple dominant eigenvalue of $\mathcal{L}_{1,c}$, with a strictly positive right eigenfunction and a positive left eigenmeasure. Its logarithm equals the pressure in [equation \(2.5\)](#).*

Proof. Use the graph coordinate of [lemma 4.1](#). A right eigenvector in the central plane can be written $(x_c, 1)$, where the second row of the eigenvalue equation gives

$$x_c = \frac{\lambda_+(c) - 1 - f}{a + g} = \frac{\kappa_b}{a} \sqrt{\delta} + o(\sqrt{\delta}) > 0. \quad (6.8)$$

Lifting to the full space gives

$$h_c = H + x_c G + O(\delta^{1-\alpha}) \quad (6.9)$$

in C^α . Choose a small fixed neighborhood of zero on which $G \geq 1/2$. There $H \geq 0$ and x_c is a positive constant times $\sqrt{\delta}$, while the error is $o(\sqrt{\delta})$ when $\alpha < 1/2$. Outside that neighborhood, H is bounded below by a positive constant. This proves $h_c > 0$ everywhere for sufficiently small $c \neq 0$.

The positive operator $\mathcal{L}_{1,c}$ has the norm limit $\lambda_+^{-n} \mathcal{L}_{1,c}^n \rightarrow \Pi_+$ on C^α for each fixed $c \neq 0$, since all its other spectrum is strictly smaller in modulus. Thus Π_+ is positive. Write $\Pi_+ f = h_c \ell_c(f)$, with $\ell_c(h_c) = 1$. Its positivity makes ℓ_c a positive functional. It is bounded for the uniform norm, because $|\ell_c(f)| \leq \ell_c(1) \|f\|_\infty$ for real f . Its extension to continuous functions is a positive finite measure. Normalize this measure to obtain $\nu_c(1) = 1$.

A change of variables gives

$$\int_0^1 \mathcal{L}_{1,c}^n 1 \, dx = b^n \int_0^1 F_{1,n,c} \, dx \leq Z_{1,n}(c). \quad (6.10)$$

The left side has positive leading coefficient times λ_+^n , so it supplies the lower pressure bound. For the upper bound, fix $c \neq 0$ and replace the weight $m_b(x - c)$ by $m_b(x - c) + \epsilon$. Its operator converges in norm to $\mathcal{L}_{1,c}$ as $\epsilon \downarrow 0$. The simple leading eigenvalue and positive eigenfunction persist, say as λ_ϵ and h_ϵ . The logarithm of this regularized weight is Hölder continuous and bounded. On each b -adic cylinder, inverse-branch contraction bounds the variation of its n -step logarithmic weight by a geometric series, independently of n . Cylinder suprema are consequently comparable, with an n -independent factor, to the weighted inverse-branch sum at any fixed target point. That sum is $\mathcal{L}_\epsilon^n 1$ and has exponential rate λ_ϵ . Hence the regularized cylinder pressure is $\log \lambda_\epsilon$. Since the original products are pointwise no larger, their upper exponential rate is at most $\log \lambda_\epsilon$. Letting $\epsilon \downarrow 0$ proves the claim. This argument needs no bounded logarithm for the original weight. $\square$

Completion of theorem 2.1. Expand the logarithm of equation (2.9) for the positive branch. The quadratic term subtracts $\kappa_b^2 \delta / 2 = d \log b \delta$. All cross terms involving $\sqrt{\delta}$ and $\delta \log \delta$, and all higher Taylor terms, are $O_\eta(\delta^{3/2-\eta})$. Thus $B_b = C_b - d \log b$, which is exactly equation (2.8). At $c = 0$, the telescoping product is $H(b^n x) / (b^n H(x))$. On a cell at distance comparable to k/b^n from an endpoint its supremum is $O(1/k)$, while the two endpoint cells contribute $O(1)$. Thus $1 \leq Z_{1,n}(0) \leq C(n+1)$, proving $P_{b,1}(0) = 0$. The expansion therefore extends continuously to zero. $\square$

Remark 6.2 (What the remainder does and does not say). For each fixed $\eta > 0$, the remainder is smaller than δ . It is not asserted to be uniform as $\eta \downarrow 0$, and no explicit numerical bound for its constant is claimed. The calculation leaves terms at the scale $\delta^{3/2} \log^2 \delta$ unresolved. A generic operator norm estimate is sufficient for both displayed coefficients, but not for a sharp subsequent term.

7 Individual eigenvectors and the critical boundary layer

7.1 Two normalizations that remain meaningful at the limit

Choose now any fixed $0 < \alpha < 1/2$. The leading eigenvalue expansion, which does not require the refined remainder on this particular space, is $\lambda_+ = 1 + \kappa_b t + o(t)$. The graph error is $O(\delta^{1-\alpha}) = o(t)$. Since graph vectors lie in the kernel of both ℓ and Ξ , the lift of $(x_c, 1)$ has $\Xi(h_c) = 1$ exactly. Equation (6.8) proves equation (2.13).

A left eigenvector of the central matrix can be written $(1, q_c)$. Its first component equation gives

$$q_c = \frac{\lambda_+(c) - 1 - e}{a + g} = \frac{\kappa_b}{a} t + o(t). \quad (7.1)$$

To lift this row to the full dual space, compose it with the coordinate map from the perturbed cluster projection. That map equals $(\ell, \Xi) + O(\delta^{1-\alpha})$ in operator norm. The resulting functional is $\ell + q_c \Xi + o(t)$. Because $\Xi(1) = 0$, its value at one is $1 + o(t)$; normalizing that value to one proves equation (2.14). Positivity and uniqueness were established in lemma 6.1.

It follows, in particular, that

$$\nu_c \implies \delta_0, \quad h_c \longrightarrow H, \quad \nu_c(h_c) = \frac{2\kappa_b}{a} t + o(t). \quad (7.2)$$

The weak convergence follows first on Hölder test functions, then on all continuous functions by uniform approximation and the fact that the ν_c are probabilities.

7.2 The boundary layer

On a fixed compact set of real z , periodicity and the expansion of sine give $H(tz)/t \rightarrow |z|$, whereas $G(tz) \rightarrow 1$ uniformly. The C^α remainder in equation (2.13) is $o(t)$ even in uniform norm. This proves equation (2.15).

The scaling is not arbitrary: the bulk eigenfunction vanishes linearly at the atom, while the positive lift at the atom is of order t . Balancing these two terms forces the spatial scale $x \asymp t$. This statement concerns the right eigenfunction; it does not assert that the left eigenmeasure has a density with a comparable pointwise profile.

7.3 Why the weak left limit is not the equilibrium limit

For any fixed $f \in C^\alpha$, multiply the two expansions and use

$$\Xi(Hf) = \int_0^1 f \, dx, \quad (7.3)$$

since $H(0) = 0$. One obtains

$$\nu_c(h_c f) = \frac{\kappa_b}{a} t \left(f(0) + \int_0^1 f \, dx \right) + o(t). \quad (7.4)$$

Divide by equation (7.2) to obtain the half-and-half limit in equation (2.17). Uniform approximation extends it to continuous test functions.

A naive multiplication of the limiting objects would give $H\delta_0 = 0$, not a probability. Both the numerator and denominator in equation (2.16) vanish at order t . Their first corrections contain the information that selects a nontrivial mixture. Also, the expansion of ν_c is in $(C^\alpha)^*$, not in the total-variation norm; the finite-part functional Ξ must not be mistaken for a signed probability correction.

Corollary 7.1 (Exact leading amplitude divergence). *For each fixed sufficiently small $c \neq 0$ there are $\theta_c < 1$ and $\mathcal{A}_b(c) > 0$ such that*

$$b^n \int_0^1 F_{1,n,c} \, dx = \mathcal{A}_b(c) e^{nP_{b,1}(c)} (1 + O_c(\theta_c^n)), \quad (7.5)$$

and

$$\mathcal{A}_b(c) \sim \frac{a}{\pi^2 \kappa_b} |c|^{-1/2}. \quad (7.6)$$

Proof. The rank-one spectral projection is $f \mapsto h_c \nu_c(f) / \nu_c(h_c)$. Integrate its value at $f = 1$ and use $\int H = 2/\pi^2$ and equation (7.2). The complementary spectrum gives the fixed- c exponential error. $\square$

The divergence order is already visible in the predecessor's spectral projectors. Formula equation (7.6) records its exact constant for the normalization equation (2.2). The two successive limits in this corollary are not a new uniform finite-size theorem.

8 Detuned eigenvectors and the selected mixture

The leading eigenvalue crossover used in this section was already proved in [4]. We give a short reconstruction that also produces the eigenvectors needed for the new selection statement.

8.1 An analytic atomic cluster

For real s close to one, put $H_s = H^s$ and $\beta_s = b^{1-s}$. The exact coboundary gives

$$\mathcal{L}_{s,0}H_s = \beta_s H_s, \quad \ell \mathcal{L}_{s,0} = \ell. \quad (8.1)$$

For fixed $\alpha < 1$, the map $s \mapsto \mathcal{L}_{s,0}$ is analytic in a complex neighborhood of one on C^α . At a simple zero, the local factor is $|x|^s$ times a nonvanishing analytic factor; the derivatives $|x|^s \log^k |x|$ have the necessary Hölder bounds uniformly whenever $\Re s > \alpha$. Away from the zeros analyticity is immediate. The same reasoning applies to H_s .

Let Π_s be the analytic rank-two cluster projection, define $G_s = \Pi_s 1$, and write its second coordinate as Ξ_s , so that

$$\Pi_s f = G_s \ell(f) + H_s \Xi_s(f). \quad (8.2)$$

Indeed $\ell \Pi_s = \ell$, because ℓ is a left eigenvector at one inside the contour. Hence $G_s(0) = 1$ and the kernel of ℓ in the central plane is generated by H_s . The coordinates satisfy

$$G_1 = G, \quad \Xi_1 = \Xi, \quad \Xi_s(H_s) = 1, \quad \Xi_s(G_s) = 0. \quad (8.3)$$

They depend analytically on s . In this basis the atomic central matrix has the form

$$A_{s,0} = \begin{pmatrix} 1 & 0 \\ a_s & \beta_s \end{pmatrix}, \quad a_s = a + O(s-1). \quad (8.4)$$

No explicit formula for a_s is needed for leading selection.

8.2 Balanced coordinates and the two eigenvectors

Set $c = \sigma t^2$ and $s = 1 + ut$. Fix $0 < \alpha < 1/2$ and let u range in a compact set. The phase perturbation has norm

$$\|\mathcal{L}_{s,c} - \mathcal{L}_{s,0}\|_{\alpha \rightarrow \alpha} = O(t^{2-2\alpha}) = o(t), \quad (8.5)$$

uniformly. For $s < 1$ use the local exponent $s - \alpha$; the extra factor $t^{2ut} = \exp(2ut \log t)$ stays bounded and tends to one. For $s \geq 1$ the uniform Lipschitz estimate suffices.

Evaluation at the atom supplies a sharper estimate for the small upper-right entry:

$$\ell((\mathcal{L}_{s,c} - \mathcal{L}_{s,0})H_s) = d|c|^s(1 + O(t^2)) = dt^2(1 + o(1)). \quad (8.6)$$

To check this, each branch j/b , $1 \leq j < b$, contributes $|c|^s(1 + O(c))$, while the zeroth branch vanishes. The formula is uniform for s near one and for either sign of c .

Compress by the graph lemma and change coordinates by $S_t = \text{diag}(t, 1)$. The square of the error in equation (8.5) is $o(t^2)$, so it remains negligible even in the small entry. Thus

$$S_t^{-1} A_{s,c} S_t = I + t \begin{pmatrix} 0 & d \\ a & -uL \end{pmatrix} + o(t). \quad (8.7)$$

Define

$$D(u) = \sqrt{u^2 L^2 / 4 + \kappa_b^2}, \quad A(u) = D(u) - uL/2, \quad B(u) = D(u) + uL/2. \quad (8.8)$$

These are positive, with $A(u)B(u) = ad$ and $A(u) + B(u) = 2D(u)$. The dominant generator eigenvalue is $A(u)$, while its right balanced vector is $(B(u)/a, 1)$ and its left balanced row is proportional to $(1, A(u)/a)$. Undoing the scaling and lifting back to the function space gives, uniformly for bounded u ,

$$\lambda_+(s, c) = 1 + tA(u) + o(t), \quad (8.9)$$

$$h_{s,c} = H_s + \frac{B(u)}{a} t G_s + o(t), \quad (8.10)$$

$$\nu_{s,c} = \ell + \frac{A(u)}{a} t \Xi_s + o(t). \quad (8.11)$$

The normalizations are $\Xi_s(h_{s,c}) = 1$ and $\nu_{s,c}(1) = 1$. For the latter, $\Xi_s(1) = O(s-1) = O(t)$, so normalization changes no first-order term. The argument for strict positivity in lemma 6.1 applies uniformly on compact u sets: $B(u)$ has a positive minimum there and the graph error is $o(t)$. The same regularization argument identifies the pressure with $\log \lambda_+$.

8.3 The mixture weights

For each $f \in C^\alpha$, $\Xi_s(H_s f) \rightarrow \Xi(Hf) = \int f \, dx$. Pairing equations (8.10) and (8.11) therefore gives

$$\nu_{s,c}(h_{s,c} f) = \frac{t}{a} \left(B(u) f(0) + A(u) \int f \, dx \right) + o(t), \quad (8.12)$$

$$\nu_{s,c}(h_{s,c}) = \frac{2D(u)}{a} t + o(t). \quad (8.13)$$

The denominator is uniformly bounded below by a positive multiple of t for bounded u . Dividing proves equation (2.18), since $w_b(u) = B(u)/(2D(u))$. Uniform approximation extends the conclusion to each continuous test function. This proves the weak-limit part of theorem 2.3, once the equilibrium interpretation is established below.

The same expansion gives the detuned right boundary layer

$$\frac{h_{1+ut, \sigma t^2}(tz)}{t} \longrightarrow |z| + \frac{B(u)}{a}, \quad (8.14)$$

uniformly for (u, z) in compact sets. In fact $H(tz)^{1+ut}/t \rightarrow |z|$, including $z = 0$, and $G_s(tz) \rightarrow 1$. Positive detuning increases the atomic lift and the selected atomic weight; negative detuning favors the Lebesgue component. These statements are about the bounded scaling window, not fixed- s extrapolations.

9 Equilibrium measures and the critical equilibrium segment

9.1 A variational argument that allows zeros

For a weight $w = m_b(\cdot - c)^s$, an equilibrium probability means a T_b -invariant probability maximizing

$$h_\mu(T_b) + \int \log w \, d\mu. \quad (9.1)$$

The value $-\infty$ is permitted for the integral. The potential is not bounded below, so this interpretation should be justified rather than attached automatically to a transfer eigenvector.

Lemma 9.1 (Positive spectral data give the unique equilibrium). *Suppose the transfer operator for a nonnegative continuous Hölder weight has a strictly positive eigenfunction h , a positive finite left eigenmeasure ν , and a simple dominant eigenvalue λ on C^α . Then $\mu = h\nu/\nu(h)$ is invariant and is the unique equilibrium probability. Its free energy equals $\log \lambda$.*

Proof. Define the normalized inverse-branch weights

$$g(y) = \frac{w(y)h(y)}{\lambda h(T_b y)}. \quad (9.2)$$

They are nonnegative and sum to one over each fiber of T_b . For continuous f , $\mathcal{L}(h(f \circ T_b)) = \lambda h f$, which, paired with ν , proves invariance of μ . More generally, for continuous F and K ,

$$\int F(y) K(T_b y) d\mu(y) = \int K(x) \sum_{T_b y=x} g(y) F(y) d\mu(x).$$

Thus the conditional distribution of a predecessor, given its image, is the finite probability vector g .

For completeness, the entropy formula needed here is

$$h_\rho(T_b) = \int \left[- \sum_{T_b y=x} p_x(y) \log p_x(y) \right] d\rho(x), \quad (9.3)$$

where p_x is the conditional predecessor distribution of any invariant probability ρ . The partition into the b half-open digit intervals is generating. In its stationary digit process, the conditional entropy of the present digit given the entire future equals the entropy rate: for finite futures it is the difference of two consecutive block entropies, by stationarity, and these differences decrease to that rate. The image point encodes the entire future, so conditioning on it gives equation (9.3). The half-open convention handles the countable ambiguity of terminating expansions and does not change the argument.

For $\rho = \mu$, equation (9.3) is at most $\log b$. Therefore $-\log g$ is μ -integrable, with $0 \log 0 = 0$. Since $\log h$ is bounded, equation (9.2) makes $\log w$ integrable as well. Its coboundary integrates to zero, giving free energy $\log \lambda$.

For arbitrary invariant ρ with finite potential integral, equations (9.2) and (9.3) express the free energy minus $\log \lambda$ as

$$- \int \sum_{T_b y=x} p_x(y) \log \frac{p_x(y)}{g(y)} d\rho(x) \leq 0. \quad (9.4)$$

The inequality is the finite-vector relative-entropy inequality, with the value $+\infty$ when positive mass is assigned to $g = 0$. A measure with potential integral $-\infty$ cannot maximize. Equality forces $p_x = g$ almost everywhere. For such a measure, the finite measure $h^{-1}\rho$ is a left eigenmeasure of $\mathcal{L}$ at λ . The adjoint eigenspace there is one-dimensional because the eigenvalue is isolated and algebraically simple on C^α . Thus the measure is proportional to ν , and normalization forces $\rho = \mu$. $\square$

The hypotheses hold at all sufficiently small nonzero phases considered in theorems 2.2 and 2.3. This completes their equilibrium assertions without invoking a positivity theorem that would fail at the defective atomic operator.

9.2 What is being selected at the critical point?

Proposition 9.2 (The critical equilibrium segment). *For $s = 1$ and $c = 0$, the equilibrium probabilities are exactly $v\delta_0 + (1 - v)\mathbf{m}$, $0 \leq v \leq 1$.*

Proof. Let ρ be ergodic and different from δ_0 . Invariance implies that the countable union of preimages of zero has ρ -measure zero. For almost every x , the telescoping relation is therefore valid:

$$\sum_{j=0}^{n-1} \log m_b(T_b^j x) = \log H(T_b^n x) - \log H(x) - n \log b. \quad (9.5)$$

There is some $\epsilon > 0$ for which $\rho(H \geq \epsilon) > 0$. Ergodicity gives infinitely many returns of almost every point to this set. Along those return times, the average in equation (9.5) tends to $-\log b$. Since $-\log m_b \geq 0$, the ergodic theorem with truncation implies that its averages tend to $+\infty$ if its integral is infinite, and to its integral otherwise. The return subsequence excludes infinity. Hence

$$\int \log m_b d\rho = -\log b. \quad (9.6)$$

Every such ergodic measure has free energy $h_\rho(T_b) - \log b \leq 0$. Equality requires entropy $\log b$. In the stationary digit process, maximal entropy means that the conditional law of each digit given its past is the uniform distribution; the digits are therefore independent and uniform. The corresponding circle measure is exactly $\mathbf{m}$.

The fixed-point measure δ_0 has both entropy and potential integral zero, so it also has free energy zero. Entropy is affine under ergodic decomposition, and the potential integral decomposes as well. Consequently a general invariant probability attains zero precisely when almost all of its ergodic components are δ_0 or $\mathbf{m}$. Their mixtures have free energy zero, which is the maximum and agrees with $P_{b,1}(0) = 0$. $\square$

This also proves the last assertion of [theorem 2.3](#). The segment classification is supplied to identify the variational meaning of the limit; the phase-selected weights and boundary behavior are the new conclusions targeted here.

10 Numerical diagnostics and reproducibility

10.1 Independent checks of the response identities

The accompanying code `verify.py` records its actual output in `data/verification.json`. At 70-decimal-digit working precision, 20 choices of base and point test both Jordan identities, and eight base–phase pairs test both the exact atom response and the bulk sine identity. The largest absolute residuals observed in these two groups are approximately 1.4×10^{-70} and 1.7×10^{-71} . These are high-precision numerical checks of independently proved identities, not interval-arithmetic certificates.

The script also evaluates the finite-part integral directly after splitting at bc . At $c = 10^{-7}$, it gives the following comparisons:

b	$\ell(E_c G)/c$	r_b	$\Xi(E_c H)/c + 2d \log(1/c)$	k_b
2	3.023854148	3.023854641	0.289459278	0.289459772
3	7.094204255	7.094205570	2.200778002	2.200779976
5	17.099438853	17.099442801	8.488157046	8.488164942

The residual differences in this table are mainly finite- c corrections, not claims of numerical error in the limiting constants. For the binary closed formula [equation \(5.12\)](#), two direct integrals with an endpoint cutoff 10^{-30} differ from the exact expression by about 10^{-30} , as expected from omitting a bounded endpoint integrand.

10.2 Pressure: a deliberately demanding observable

For the continuum pressure define the logarithm-corrected quotient

$$R_b(c) = \frac{P_{b,1}(c) - \kappa_b \sqrt{|c|} + (b-1)|c| \log(1/|c|)}{|c|}. \quad (10.1)$$

The theorem predicts $R_b(c) \rightarrow B_b$. The code discretizes the transfer operator by nonnegative periodic piecewise-linear interpolation at the inverse branches and computes its two eigenvalues with largest real parts. The following are floating-point collocation values, not certified continuum values:

b	c	$P_{b,1}(c)/\sqrt{ c }$	$R_b(c)$	B_b
2	10^{-3}	1.02347259	2.039826	0.963510
2	10^{-4}	1.10259410	1.728748	0.963510
2	10^{-5}	1.14540630	1.392461	0.963510
2	10^{-6}	1.16477281	1.178294	0.963510
3	10^{-3}	1.80542336	4.617369	2.450268
3	10^{-4}	1.95254224	4.045490	2.450268
3	10^{-5}	2.03409965	3.358224	2.450268
3	10^{-6}	2.07156876	2.905634	2.450268

The leading constants are $\kappa_2 \approx 1.17741002$ and $\kappa_3 \approx 2.09629415$. The displayed approach of $R_b(c)$ to B_b is slow. That is compatible with the remainder and with additional $|c|^{3/2} \log^2 |c|$ -scale contributions; the table does not establish a coefficient for any such term.

The nominal meshes are 65 536 points at $c = 10^{-3}$, 262 144 at 10^{-4} and 10^{-5} , and 1 048 576 at 10^{-6} . They are rounded upward to a multiple of the base. A doubled-grid check for $b = 2$, $c = 10^{-6}$ changes the pressure from 0.00116477280617 to 0.00116476829537. The change is approximately 4.51×10^{-9} in pressure but about 0.00451 after division by c in equation (10.1). This illustrates why a small eigenvector residual alone does not certify the continuum coefficient. Both meshes still give values above the predicted limiting constant, consistent with nonnegligible finite-phase corrections.

10.3 Selection: an observable distinguishing the two components

The test function $\cos 2\pi x$ has value one at the atom and integral zero against m . Hence its predicted limiting expectation is exactly $w_b(u)$. Computing left and right eigenvectors of the same positive collocation matrix and normalizing their entrywise product gives:

u	c	numerical $\int \cos(2\pi x) d\mu$	predicted limit
-2	10^{-4}	0.26869569	0.24633959
-2	10^{-5}	0.25711076	0.24633959
0	10^{-4}	0.51963407	0.50000000
0	10^{-5}	0.50961315	0.50000000
2	10^{-4}	0.77612951	0.75366041
2	10^{-5}	0.76440084	0.75366041

All these rows use $b = 2$ and 262 144 mesh points. The recorded output also includes the sine moment, matrix residuals, and the discrepancy between the eigenvalues obtained from left and right calculations. The agreement is qualitative and improving across the two phases; it is not a rigorous weak-convergence certificate. No numerical result is used as a premise in any theorem above.

10.4 How to reproduce the package

With the versions recorded in `requirements.txt`, run

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python code/verify.py
```

The environment-variable syntax shown is for a POSIX shell; it only limits numerical threading. On other shells set it using their syntax or omit it. The script records library versions and computation residuals. To skip collocation and retain only the special-function diagnostics, use `python code/verify.py -identities-only`. The option `-output-dir` sends outputs to a different directory so the shipped receipt need not be overwritten. The PDF is rebuilt by the supplied `Makefile`, or by three successive runs of pdfLaTeX. No network access is needed once the dependencies are installed.

11 Further research questions and concrete next targets

The questions below are proposed continuations of this article. They are not all claimed to be previously published open problems. The first, fourth, and fifth also refine explicitly identified boundaries in the critical predecessor.

Research question 1 (The next critical logarithmic polynomial). Does the next pressure term have the form $|c|^{3/2}Q_b(\log |c|)$, with Q_b a polynomial of degree two? The square of the diagonal difference in equation (6.3) already creates such a candidate, but the other matrix entry and the graph correction may contribute at the same scale. Compute those contributions before assigning a coefficient. The binary exact formula equation (5.12) isolates one part of the task.

Research question 2 (A sharp remainder without a power loss). Replace $O_\eta(|c|^{3/2-\eta})$ by a bound involving explicit powers of $\log(1/|c|)$ and, ideally, computable constants. The obstruction is the use of a global Hölder norm in PEK . A localized estimate pairing E_c with the reduced resolvent may keep track of the small support of the cusp displacement instead of paying $|c|^{-\alpha}$ twice.

Research question 3 (A complete critical expansion). Determine whether the pressure and its central spectral data have an asymptotic expansion in powers of $\sqrt{|c|}$ and $\log |c|$. If so, give a recursive algorithm and prove closure of the relevant function class under the reduced-resolvent operations. The present three-term formula excludes a pure Puiseux series but does not exclude additional log-periodic or other nonalgebraic contributions at later orders.

Research question 4 (Growing detuning and matched selection). Extend equation (2.18) to $|u| \leq U(t)$ with $U(t) \rightarrow \infty$. The factors t^{2ut} and the shrinking relative central gap must be controlled together. Determine a quantitative overlap with fixed supercritical and subcritical regimes, for both pressure and equilibrium weights. Taking $u \rightarrow \infty$ in the limiting formula alone does not establish that theorem.

Research question 5 (Two-point regularity and differentiated remainders). Is the pressure $C^{0,1/2}$ on a full neighborhood of zero? Can its one-sided derivative, away from exceptional nonzero phases, be expanded by differentiating equation (2.8)? An asymptotic remainder in function values does not justify differentiating it. Control between two moving phases, not just between a phase and zero, is required.

Research question 6 (Quantitative convergence of equilibrium measures). Obtain explicit rates for the convergence to the selected mixtures, first in the dual of a fixed Hölder space and then in a transport metric. Determine whether the atomic weight admits a $\sqrt{|c|} \log(1/|c|)$ correction and whether its coefficient can be read from the refined eigenvalues plus localized eigenvector estimates. This would quantify the finite-phase bias visible in the cosine table.

Research question 7 (The left boundary layer). The right eigenfunction has the explicit profile equation (8.14). Determine how the probability $\nu_{s,c}$ concentrates near zero and whether rescaling its restriction yields a nontrivial measure. The first dual correction Ξ is a finite-part functional rather than a measure, so it cannot itself be used as a probability profile. A useful formulation should separate concentrated mass from the small diffuse tail.

Research question 8 (Other masks and other Jordan multiplicities). For which nonconsecutive digital masks do a defective isolated spectral cluster and a finite-part dual functional occur? Establish conditions under which a logarithmic trace response is universal. For a Jordan chain of length greater than two, determine the analogous root scaling, equilibrium simplex, and selection rule. A finite matrix prediction must be supported by an actual operator realization.

Research question 9 (Certified computation and proof-assistant integration). Turn the weighted quadrature and graph-compression estimates into explicit constants, then produce interval enclosures for a nontrivial phase neighborhood. A proof-assistant route can separate the exact cotangent

and digamma identities, the endpoint integral estimates, the finite matrix calculation, and the spectral/entropy bridges. The exact finite-dimensional algebra is a useful first target, but formalizing only that calculation would not verify the infinite-dimensional theorem.

A Proof dependencies and an audit of scope

The pressure proof follows the dependency chain

telescoping quotient and weighted quadrature $\implies$ isolated normalized Jordan block
 $\implies$ graph compression with a quadratic error
 $\implies$ exact atom/bulk response and finite-part logarithm
 $\implies$ refined eigenvalues $\implies$ pressure via positive regularization.

The equilibrium proof additionally uses the first left/right eigenvector corrections, the cancellation $\Xi(Hf) = \int f$, and the conditional entropy argument. It does not follow just from the leading pressure asymptotic or from a pointwise limit of the right eigenfunction.

Three verification boundaries should remain explicit. First, the present manuscript is not a Lean or Rocq development. Second, collocation output is not an enclosure of the spectrum of the continuous operator. Third, a theorem proved in a new research draft is not the same thing as an independently established priority claim. The source ledger records the specific repository questions addressed, and the manuscript reconstructs the analytic ingredients rather than importing unreviewed claims as unlabeled axioms.

Claim	Verification supplied
Normalized Jordan chain and gap	Written proof; high-precision checks of the exact identities.
Logarithmic and linear pressure coefficients	Written finite-part and matrix proof; independent integral and collocation diagnostics.
Boundary layer and mixture selection	Written norm/duality argument; left/right collocation checks of a distinguishing observable.
Equilibrium meaning and uniqueness	Written conditional-entropy proof, including zeros of the weight.
PDF and package integrity	Compilation, reference and font checks, page rendering, and a build-validation receipt.

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- [4] ProveIt research corpus, research draft prepared with ChatGPT for V. Reshetnikov, *Critical Cusps in Digital-Product Pressure: A Jordan collision, square-root phase scaling, and explicit subcritical response*, 29 September 2026. At the snapshot in [3], source `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Critical_Pressure/article.tex`. [Pinned source](#). The research questions “The next critical term” and “Eigenfunctions, eigenmeasures, and equilibrium measures” are the immediate targets.
- [5] ProveIt research corpus, research draft prepared with ChatGPT for V. Reshetnikov, *Fractional Cusps at the Atomic Phase: Sharp noninteger regularity of Thue–Morse pressure and sharp relaxation for digital products in every base*, 29 September 2026. At the snapshot in [3], source `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Fractional_Pressure/article.tex`. [Pinned source](#).
- [6] ProveIt research corpus, *Thue_Morse_Subcritical_Pressure*, research package, 29 September 2026. At the snapshot in [3], source `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Subcritical_Pressure/article.tex`. [Pinned source](#). Its all-subcritical extension is also recorded in the editorial note in [4]; no theorem from this package is assumed here.